SWIFT SHIP TRACKER

User Story:

The SwiftShip Tracker is designed to provide a centralized Salesforce-based parcel management solution that enables customers to book parcels, track delivery status, and manage shipments through an intuitive interface and conversational AI support. The system manages Parcel, Delivery, Sender, and Receiver information using Salesforce custom objects and relationships. Flows automate parcel operations and notifications, while AgentForce AI and Prompt Builder enable customers to retrieve parcel information and interact with the system conversationally. Delivery agents can update parcel status in real time, while Experience Cloud, security controls, reports, and dashboards provide secure customer access and operational visibility.

1. INTRODUCTION

1.1Project Overview :

The SwiftShip Tracker is a Salesforce CRM solution developed to simplify parcel booking, shipment tracking, and delivery management. It supports the complete parcel lifecycle from booking and dispatch through tracking and delivery confirmation. Customers can access shipment information, while delivery agents can update parcel details and delivery status.

1.2 Purpose

- Centralize parcel and delivery information.
- Simplify parcel booking and tracking.
- Provide real-time shipment visibility.
- Automate delivery updates and customer notifications.
- Enable conversational parcel tracking using AgentForce AI and Prompt Builder.

● Improve agent efficiency through Salesforce Flows and dashboards. ● Provide secure, role-based access to customers, agents, support staff, and administrators.

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

Logistics companies and e-commerce platforms face challenges in providing customers with real-time, transparent shipment tracking. Currently, many systems rely on fragmented updates, manual entries, or third-party trackers, which lead to delays, miscommunication, and poor customer experience.

Customers demand **instant visibility** into their parcel's journey — from dispatch to delivery — while managers need **centralized dashboards** to monitor performance, detect delays, and optimize routes.

The absence of a unified, automated solution results in:

- Inefficient communication between courier agents and customers
- Lack of real-time shipment status updates
- Difficulty in analyzing delivery performance and bottlenecks
- Reduced customer trust and satisfaction

Swift Ship Tracker aims to solve this problem by integrating Salesforce CRM with a Swiftbased mobile application.

2.2 Empathy Map Canvas

User Persona: Customer / Sender

● Think & Feel: Wants a simple way to book and track parcels without repeatedly contacting support.

● Hear: Receives delivery information and status updates from the parcel service. ● See: Parcel details, tracking status, estimated delivery date, and delivery information.

Say & Do: Books parcels, checks tracking information, and asks for delivery updates.

● Pain: Delays, confusion, lack of easy tracking, and dependence on support or multiple apps.

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- Gain: Real-time visibility, self-service tracking, convenient booking, and timely notifications.



2.3 Brainstorming & Idea Prioritization

Ideas Considered

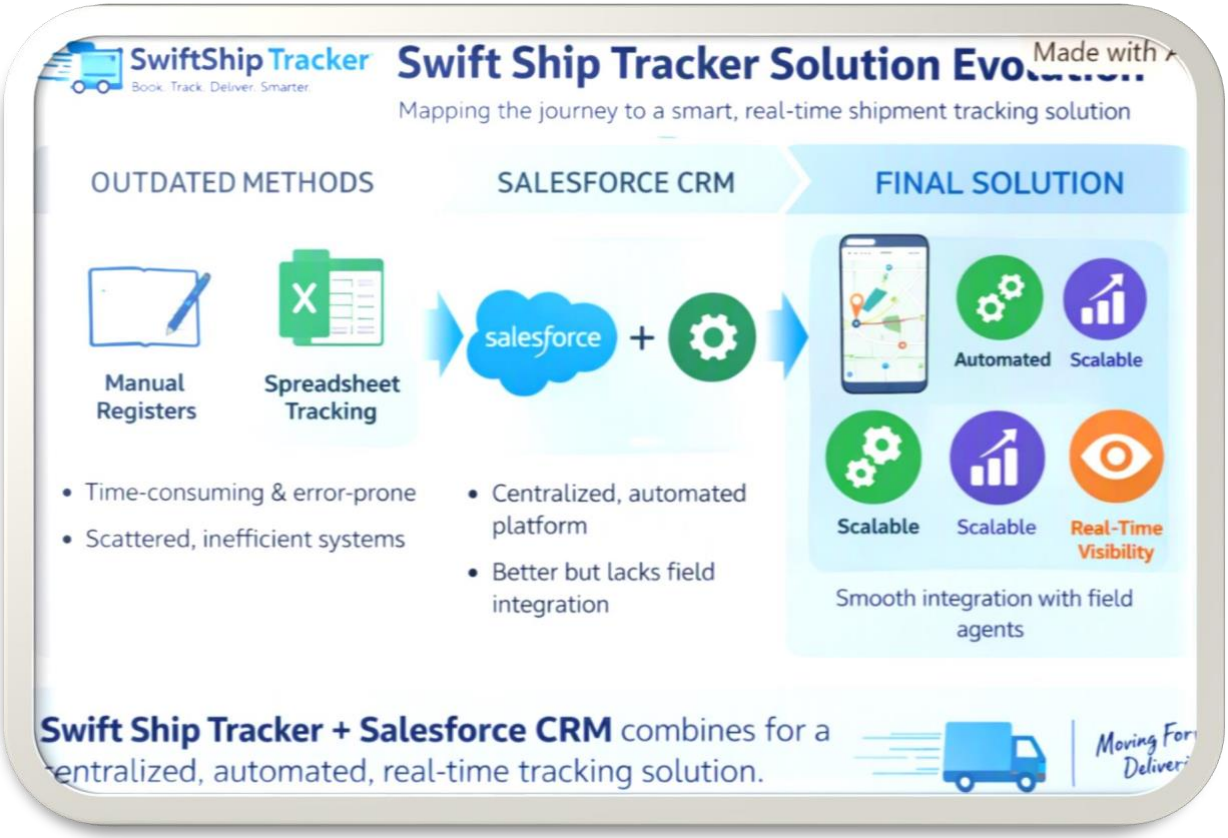
- Manual registers
- Spreadsheet tracking
- Standalone parcel portal

Salesforce CRM with AgentForce AI

Final Selection

Salesforce CRM selected due to:

- Centralized parcel management
- Automation through Flows
- Real-time tracking visibility
- Conversational AI assistance
- Secure role-based access
- Reports and dashboards



3.REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stages:

Parcel Booking

- Parcel Record Creation
- Dispatch & Shipment

- In-Transit Tracking
- Out for Delivery
- Delivery Confirmation
- Customer Notification
- Feedback / Support



3.2 SOLUTION REQUIREMENT

FUNCTIONAL REQUIREMENTS

1. User & Access Management

- FR-01: Role-based login - Admin, Operations, Delivery Agent, Customer (Sender/Receiver)

- FR-02: Self-registration for customers via Experience Cloud portal

- FR-03: OWD Private + Role Hierarchy + Sharing Rules for all custom objects

2.Parcel Lifecycle Management

- FR-04: Create Sender*c, Receiver*c with validation (mobile, pincode, GST)

- FR-05: Create Parcel__c with auto-generated unique tracking ID - format: `SWT-{YYYY}-{00000}`

- FR-06: Support bulk parcel upload via CSV (for SMB sellers)

- FR-07: Status flow: `Booked > Picked Up > In-Transit > Out for Delivery > Delivered > Exception`

- FR-08: No manual status change allowed - only via Flow / Apex trigger / Agent delivery app

3.Dispatch & Verification

- FR-09: Auto-generate digital receipt with QR code + unique number

- FR-10: Send receipt instantly to sender & receiver via Email + WhatsApp

- FR-11: QR verification to prevent fake receipt fraud

4. Real-Time Tracking (Core)

- FR-12: Branded tracking page - search by tracking ID / mobile / OTP

- FR-13: Live milestone timeline with ETA, current location, delay reason

- FR-14: Map view for parcel movement

- FR-15: Conversational AI tracking - user asks "Where is my parcel SWT-123?" - Agentforce / Einstein Bot replies with live status

5. Delivery Management

- FR-16: Delivery__c object - OTP based delivery confirmation

- FR-17: Photo proof of delivery upload by agent

- FR-18: Exception handling - Return to origin, reschedule, supervisor approval flow

6. Notifications

- FR-19: Automated notifications via Flow for every milestone - WhatsApp, SMS, Email

- FR-20: Proactive delay alerts + revised ETA

- FR-21: WISMO deflection - tracking link in every message

7. Reporting & Admin

- FR-22: Dashboards - Daily bookings, On-time delivery %, Agent performance, Exception rate, Revenue

- FR-23: Reports - Parcel by status, City-wise volume, Average transit time

- FR-24: AI insights - Predict delivery delay, identify fraud patterns

NON-FUNCTIONAL REQUIREMENTS

Performance

- NFR-01: Tracking page load < 2 sec even with 10k concurrent users

- NFR-02: Notification delivery < 30 sec from status change

- NFR-03: Support 1 Lakh parcels / month, scalable to 10 Lakh

Security

- NFR-04: All custom objects OWD Private, field-level security for PII

- NFR-05: OTP verification mandatory for delivery, tracking page rate-limited

- NFR-06: Audit trail - who changed what, when (Field History Tracking on Parcel__c)

Availability & Reliability

- NFR-07: 99.5% uptime for tracking portal - hosted on Salesforce Experience Cloud

- NFR-08: Offline capability for delivery app - sync when back online

- NFR-09: Data backup daily, retention 7 years for compliance

Usability

- NFR-10: Mobile-first - Works on WhatsApp without app download (Chennai SMB use case)

- NFR-11: Multi-language - English, Tamil, Hindi for tracking updates

- NFR-12: WCAG accessible tracking page

3.3 Data Flow Diagram

Customer / Sender

↓

Parcel Booking

↓

Parcel Record

↓

Sender + Receiver Information

↓

Delivery Record

↓

Parcel Status Update

↓

Current Location & Estimated Delivery

↓

Automated Notifications

↓

Customer Tracking / AgentForce AI

↓

Reports & Dashboards

AI TRACKING FLOW

Customer Query

↓

AgentForce AI

↓

Prompt Builder – Retrieve Parcel Details

↓

Parcel Record

↓

Status / Weight / Estimated Delivery

↓

Conversational Response

The Prompt Builder implementation retrieves Parcel ID, Status, Weight, and Estimated Delivery Date information for tracking responses.

3.4 Technology Stack

Layer	Technology
Platform	Salesforce Developer Edition
UI	Lightning App Page, Lightning Record Pages, Experience Cloud
Database	Salesforce Custom & Standard Objects
Business Logic	Salesforce Flows, Apex Classes
Automation	Record-Triggered Flows, Auto-Launched Flows, Scheduled Flows
AI	AgentForce AI, Prompt Builder
Notifications	Email Alerts & Email Templates
Processing	Batch Apex
Security	Profiles, Permission Sets, Role Hierarchy, Field-Level Security, Sharing Rules
Reporting	Reports & Dashboards

PROJECT DESIGN PHASE

Problem–Solution Fit

1. HIGH LEVEL ARCHITECTURE (HLD)

System Stack:

- Frontend: Experience Cloud (Customer Portal + Tracking Page) + Delivery Agent Mobile App

- Backend: Salesforce Platform - Custom Objects, Flows, Apex, Agentforce

- Integrations: WhatsApp Business API, SMS Gateway, Google Maps API, Payment Gateway

- Data: Salesforce Org (OWD Private)

Flow: Customer books on Portal → Flow creates Parcel__c + Tracking ID → Trigger sends receipt via WhatsApp → Agent picks → Status updates via App → Real-time notification → OTP Delivery → Dashboard updates

2.DATA MODEL DESIGN (LLD)

Core Objects:

- *Sender__c* (Master) - Name, Mobile, Email, GSTIN, Address, City, Pincode

- *Receiver__c* (Master) - Name, Mobile, Address, Landmark, Pincode, OTP

- *Parcel*c* (Central) - Tracking_ID*c (Unique), Sender*c (Lookup), Receiver*c (Lookup), Weight*c, Service_Type*c, Status*c (Picklist), ETA*c, Current_Location__c

- *Delivery*c* (Child of Parcel*c) - Agent*c, Delivery_OTP*c, Photo_Proof*c, Signature*c, Actual_Delivery_Date*c, Exception_Reason*c

- *Agent__c* - Agent ID, Zone, Mobile

Relationships:

Sender (1) → (M) Parcel (M) ← (1) Receiver

Parcel (1) → (M) Delivery attempts

Enable Field History Tracking on Parcel*c.Status*c

3.PROCESS DESIGN

A. Record-Triggered Flows (Replace Code where possible):*

- Flow-01 On Parcel__c Create: Generate Tracking ID `SWT-{YYYY}-{AutoNumber}`

- Flow-02 On Status Change: Send WhatsApp Template + Email

- Flow-03 On ETA Breach: Create Task for Supervisor + Alert

B. Apex Triggers (Only where Flow can't):*

- Trigger on Parcel__c: Prevent manual status rollback, calculate transit time

- Trigger on Delivery__c: Validate OTP, close Parcel to Delivered

C. Approval Process:

- Exception Approval: If Delivery failed >2 times → Supervisor approval for Return to Origin

D. Agentforce Design:

- Prompt Template: "You are SwiftShip assistant. Given tracking ID {Tracking_ID*c} and status {Status*c}, answer in 2 lines + share ETA"

- Data Cloud: Feed Parcel history for prediction

4. SECURITY DESIGN

- OWD: Private for all 4 objects

- Roles: Admin > Operations Manager > Zone Manager > Delivery Agent > Customer

- Sharing Rule: Sender can see only his Parcels, Agent sees only his Zone Parcels

- Profile: Customer Community profile - Read-only Parcel, No delete

5.UI/UX DESIGN

- *Tracking Page:* Single search box, Timeline component (SLDS), Map LWC

- *Booking Page:* 3-step wizard - Sender > Receiver > Parcel details

- *Agent App:* Mobile LWC - My Deliveries, OTP entry, Camera upload

- *Admin Dashboard:* 4 dashboards - Volume, SLA breach, Agent performance, Revenue

6. INTEGRATION DESIGN

Integration	Direction	Method
WhatsApp	Salesforce → Customer	HTTP Callout via Flow
SMS	Salesforce → Customer	SMS Gateway API
Maps	Salesforce ←→ Maps	LWC + Google Maps JS

7. DESIGN DELIVERABLES CHECKLIST

You must exit design phase with:

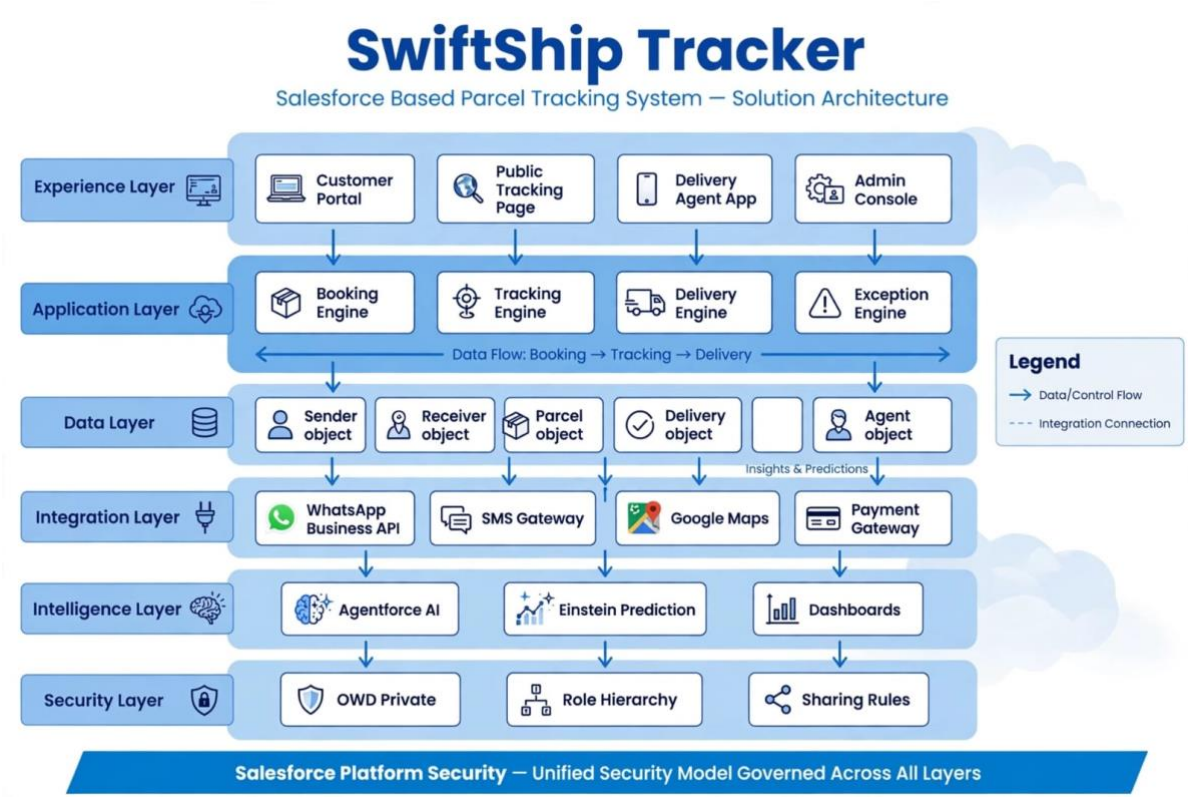
1. HLD diagram
2. ER diagram + Object schema
3. Flow diagrams for Booking → Delivery
4. Security matrix (Who can see what)
5. Prototype of tracking page in Figma / LWC
6. Test scenarios - Happy path, OTP failure, Duplicate booking

Proposed Solution

Functional Requirements

Parameter	Description
Problem	Difficult and inefficient parcel booking and tracking
Solution	Salesforce-based Parcel Management System
Innovation	AgentForce AI + Prompt Builder for conversational self-service tracking and automated Salesforce Flows
Impact	Centralized booking, real-time tracking visibility, timely notifications, and efficient delivery management
Business Model	Parcel, customer, delivery-agent, support, and administrator operations
Scalability	Supports growing parcel operations, Experience Cloud, and integration with external logistics systems

Architecture :



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Here is Project Planning and Scheduling for SwiftShip Tracker - no serial numbers.

PROJECT TIMELINE - TWELVE WEEKS*

Phase - Discovery and Requirement Finalization

Duration - One week

Activities

- Stakeholder interviews with operations and delivery agents

- Finalize BRD with functional and non functional requirements

- Sign off on tracking page prototype

Deliverable - BRD and Figma prototype

Phase - Solution Design

Duration - Two weeks

Activities

- HLD and LLD with data model for Sender Receiver Parcel Delivery Agent

- Security design with OWD Private and sharing rules

- Flow and Apex design for tracking ID and OTP validation

- Integration design for WhatsApp SMS Maps

Deliverable - Design Document and ER Diagram

Phase - Build Sprint One - Core Data and Booking

Duration - Two weeks

Activities

- Create custom objects with fields and validation rules

- Build booking wizard Screen Flow

- Auto generation of tracking ID via Flow

- Digital receipt with QR code

Milestone - Booking demo ready

Phase - Build Sprint Two - Tracking and Notifications*

Duration - Two weeks

Activities

- Public tracking page LWC with timeline and map

- Record triggered Flows for milestone notifications

- WhatsApp Business API integration via HTTP callout

- Agentforce prompt for conversational tracking

Milestone - End to end tracking live in sandbox

Phase - Build Sprint Three - Delivery and Exceptions

Duration - Two weeks

Activities

- Delivery Agent mobile LWC with OTP and photo proof

- Apex trigger for OTP validation and status closure

- Exception handling and supervisor approval process

- Offline sync capability for agent app

Milestone - Last mile ready

Phase - Testing

Duration - One week

Activities

- Unit testing with seventy five percent coverage

- UAT with operations team using real parcels in Chennai zone

- Security testing for sharing rules and OTP bypass

- Performance testing for tracking page under load

Deliverable - Test report and bug fixes

Phase - Deployment and Training

Duration - One week

Activities

- Deployment to production using change sets

- Data migration for existing senders and agents

- Training for admin operations and delivery agents

- WhatsApp template approval

Milestone - Go Live

Phase - Hypercare and Handover

Duration - One week

Activities

- Monitor SLA breach and notification failures

- Support for first week of live parcels

- Handover of admin guide and runbook

Deliverable - Project closure

RESOURCE PLAN

- Salesforce Admin / Developer for objects and Flows

- LWC Developer for tracking page and agent app

- Integration specialist for WhatsApp and Maps

- QA for UAT

- Product Owner from operations for sign off

DEPENDENCIES

- WhatsApp Business API approval before Build Sprint Two

- Maps API key before tracking page

- Agent mobile devices for UAT before Testing

GANTT VIEW

Phase Week One Week Two-Three Week Four-Five Week Six-Seven Week Eight-Nine Week Ten Week Eleven-Twelve

Discovery Active

Design Active

Build One Active

Build Two Active

Build Three Active

Testing Active

6. Project Development Phase:

Phase 1: Requirement Analysis & Planning

DEVELOPMENT APPROACH

Agile with Flow first approach. No code unless Flow cannot do it. Build in sandbox, test in partial, deploy to prod.

ENVIRONMENT SETUP

- Dev Sandbox for each developer
- Integration Sandbox for Flows and LWC integration
- UAT Sandbox with Chennai zone real data
- Production org with Experience Cloud site enabled

DEVELOPMENT TRACKS

Data Development

- Create Sender Receiver Parcel Delivery Agent objects
- Add fields with proper data types and help text
- Add validation rules for mobile format and pincode
- Enable field history tracking on Parcel Status
- Set OWD Private and create role hierarchy

Automation Development

- Screen Flow for booking wizard with three steps Sender Receiver Parcel
- Record triggered Flow for Tracking ID generation using formula `SWT-YEAR-Autonumber`
- Record triggered Flow for status change notification via WhatsApp and Email
- Scheduled Flow for ETA breach check every hour
- Apex Trigger only for OTP validation and preventing status rollback
- Approval Process for Return to Origin after two failed attempts

UI Development

- LWC component TrackingTimeline for public tracking page with SLDS timeline
- LWC component BookingWizard for portal
- LWC component AgentDeliveryList for mobile with OTP input and camera upload
- LWC component LiveMap using Google Maps JS for current location
- Experience Cloud site branding with logo and tracking search box

Integration Development

- Named Credential for WhatsApp Business API
- HTTP callout from Flow using invocable Apex for WhatsApp template
- SMS fallback via SMS gateway API
- Maps integration via static resource

AI Development

- Prompt Template for Agentforce to answer tracking queries in friendly tone
- Data mapping for Parcel history to generate delay prediction

CODING STANDARDS

- Flow naming convention - `SwiftShip_Parcel_OnCreate_TrackingID`
- Apex class naming - `SwiftShip_OTP_Validator`
- Bulkified triggers with no SOQL inside loop
- Seventy five percent test coverage with positive and negative cases
- No hardcoded IDs, use custom metadata for WhatsApp template IDs

VERSION CONTROL

- Use VS Code with SFDX
- Commit every Flow and LWC change to Git
- Pull request review before merging to integration

DEVELOPMENT CHECKLIST FOR EACH STORY

- Build in dev sandbox
- Test with bulk data for one hundred parcels
- Check sharing - Sender sees only his parcels
- Check mobile view for tracking page
- Move to integration and run regression

COMMON BUILDS IN THIS PHASE

- Booking Screen Flow demo with auto tracking ID
- Tracking page with timeline and map working in community
- WhatsApp notification received on milestone change
- Agent able to deliver with OTP and photo proof
- Dashboard showing live parcel count

EXIT CRITERIA

- All objects and Flows built and working in integration sandbox
- No critical bugs for booking to delivery happy path
- UAT ready for operations team

Want me to give you the exact Flow steps and Apex trigger code for Tracking ID generation and OTP validation?What Is Salesforce?

Salesforce is your customer success platform, designed to help you sell, service, market, analyse, and connect with your customers.

Creating Developer Account

Creating a developer org in salesforce.

Go to https://developer.salesforce.com/signup

On the sign-up form, enter the following details:

Build for the Agentic Enterprise. Sign up for your Developer Edition.

The entire Salesforce platform - exposed as APIs, MCP tools, and CLI commands. Build apps, ship agents, and let coding agents do the heavy lifting with Salesforce Headless 360.

Sign up for your Developer Edition

A free Salesforce Platform environment with Agentforce and Data 360

First name

Harish Ragavendhira .s

Last name

Abi.A,Abhinaya.A,Dhesuki.R,Pranesh.A

Job title

Salesforce Developer

Company

ARJ college of engineering and technology

Country/region

India

Work email

harishragavendhira5@gmail.com

I agree to the Main Services Agreement – Developer Services, Salesforce Program Agreement, and API Terms of Service

Your free trial may be provisioned on or migrated to Hyperforce, Salesforce's public cloud infrastructure.

I acknowledge, as described in the Developer Documentation: (1) the Developer Edition includes autonomous and other generative AI features; and (2) Salesforce may limit use of those features and the org, and may terminate any org that has been inactive for 45 days.

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Sign me up

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1) First name & Last name

2) Email

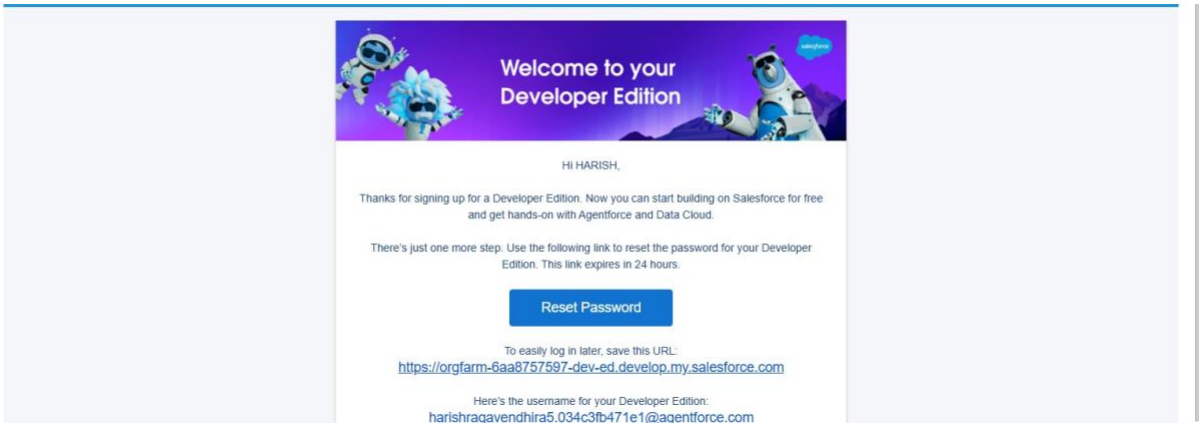
3) Job Title: Developer

4) Company: College Name

5) Country: India

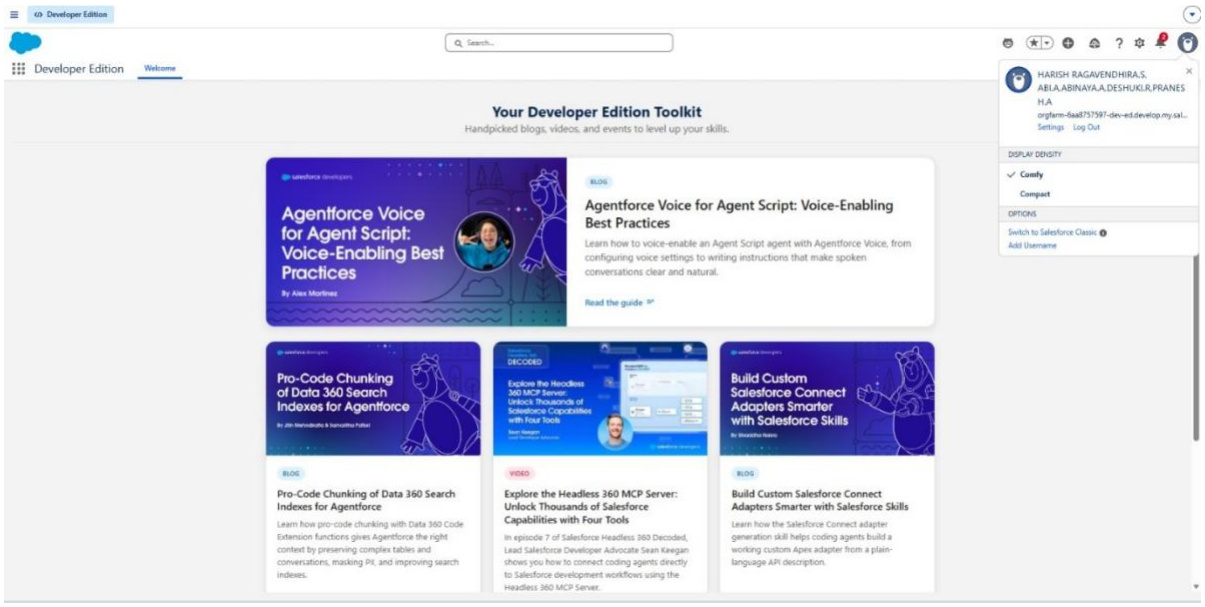
Click on sign me up after filling these.

2.Go to the inbox of the email that you used while signing up. Click on the verify account to activate your account. The email may take 10-30mins and sometimes 2 hours.



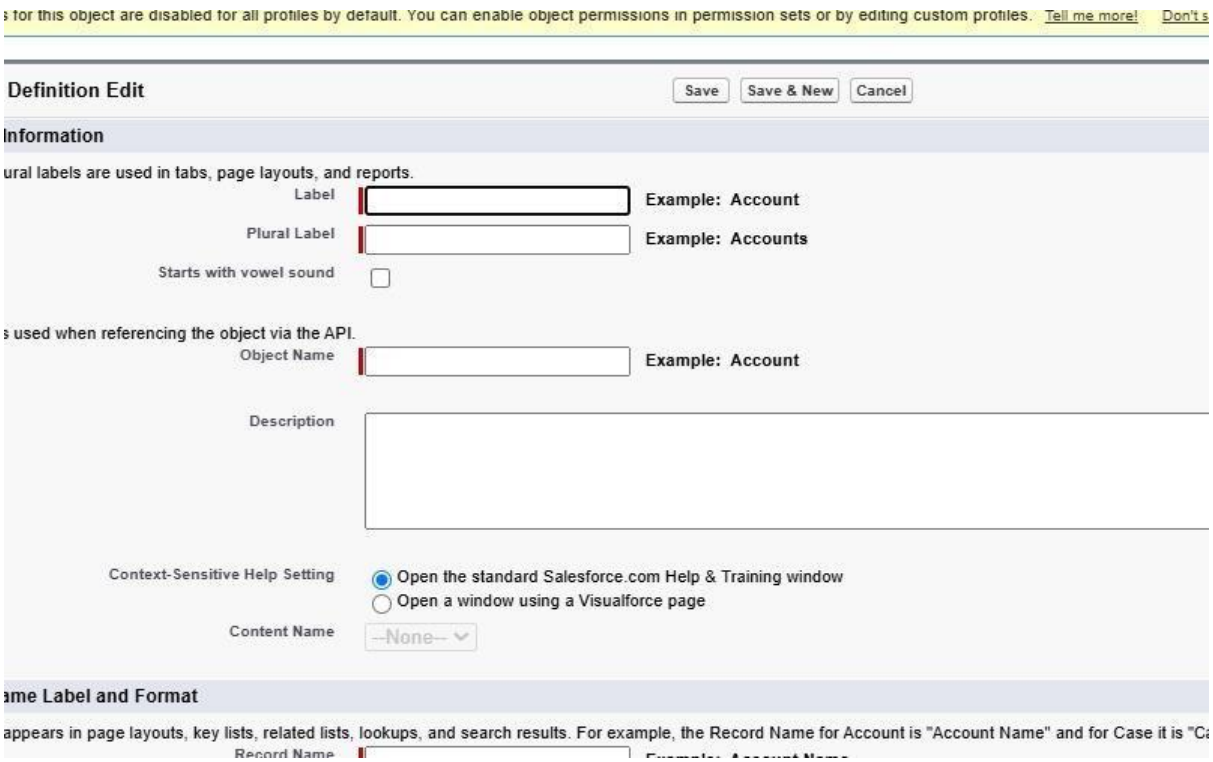
Click on Verify Account

1. Give a password and answer a security question and click on change password.
2. Then you will redirect to your salesforce setup page.



Milestone 2- Objects Creation To create an object:

1. From the setup page
2. Click on Object Manager
3. Click on Create >> Click on Custom Object.



4. Enter the label name as Parcel
5. Enter Plural label name as Parcels

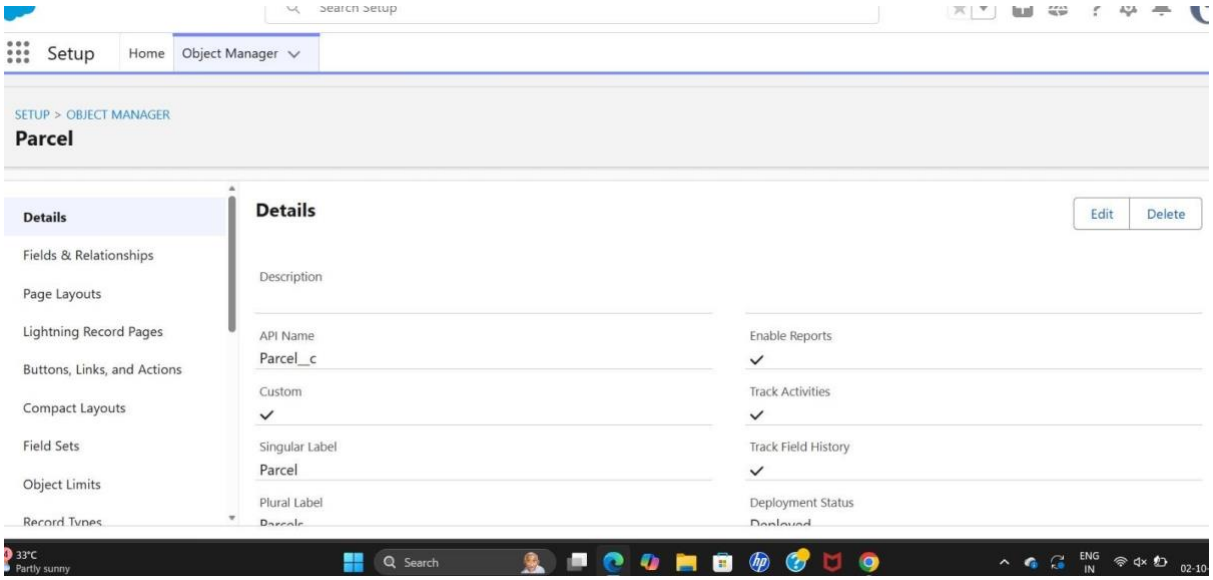
6. Enter Record Name as Parcel Name 7. Select Data Type as Text: Parcel Name

8. Select Allow reports.

9. Select Allow search.

10. Allow Track Field History

11. Click on Save and New



Creating Delivery Object To

create an object:

1. From the setup page

2. Click on Object Manager

3. Click on Create >> Click on Custom Object.

4. Enter the label name as Delivery

5. Enter Plural label name as Deliveries

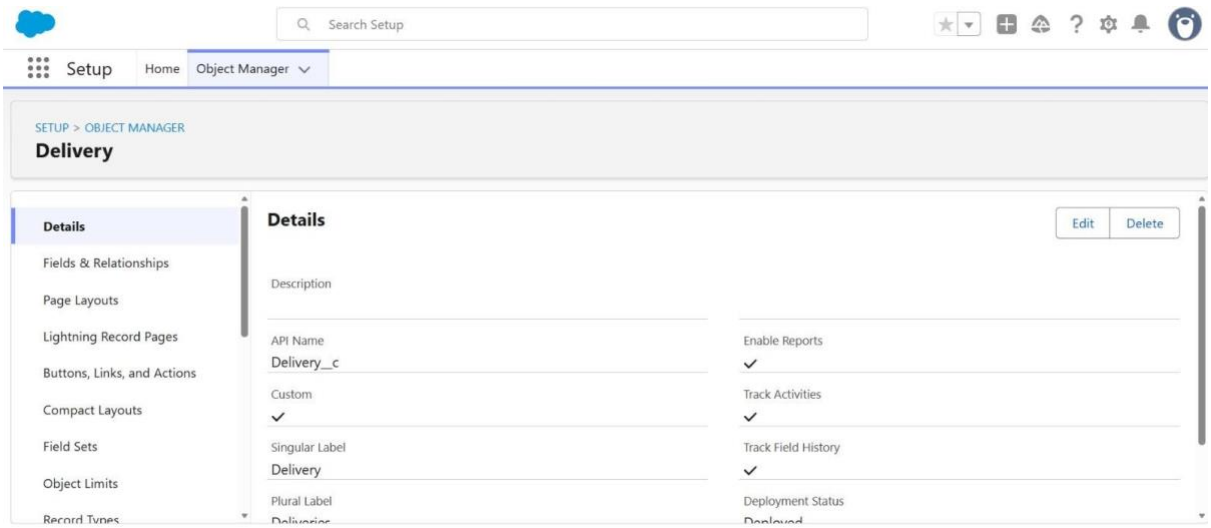
6. Enter Record Name as DeliveryName 7. Select Data Type as Text: DeliveryName

8. Select Allow reports.

9. Select Allow search.

10. Allow Track Field History

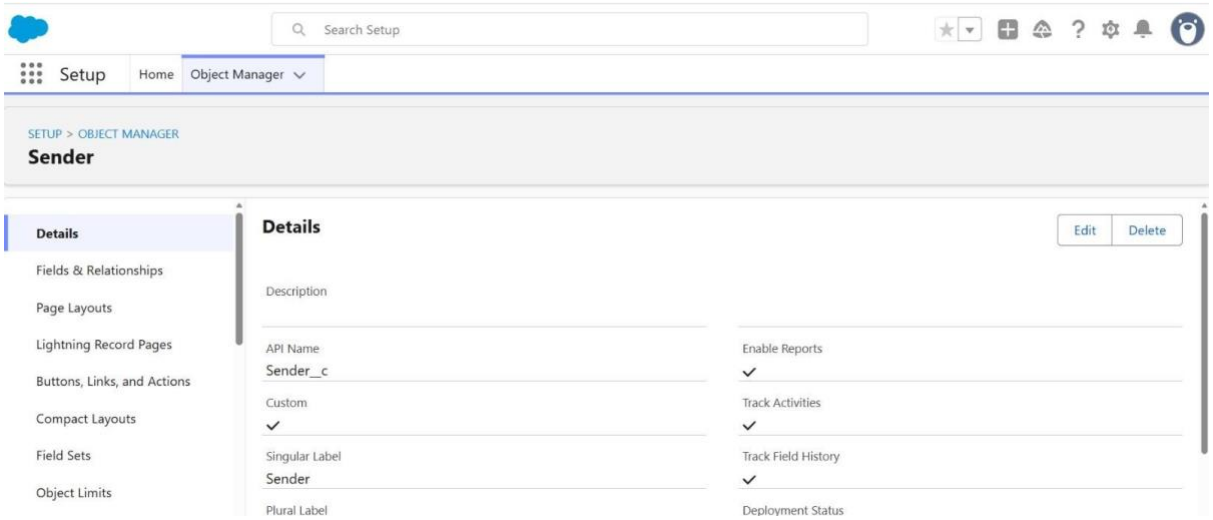
11. Click on Save and New



Creating Sender Object To

create an object:

1. From the setup page
2. Click on Object Manager
3. Click on Create >> Click on Custom Object.
4. Enter the label name as Sender
5. Enter Plural label name as Sender's
6. Enter Record Name as SenderName 7. Select Data Type as Text: SenderName
8. Select Allow reports.
9. Select Allow search.
10. Allow Track Field History
11. Click on Save and New



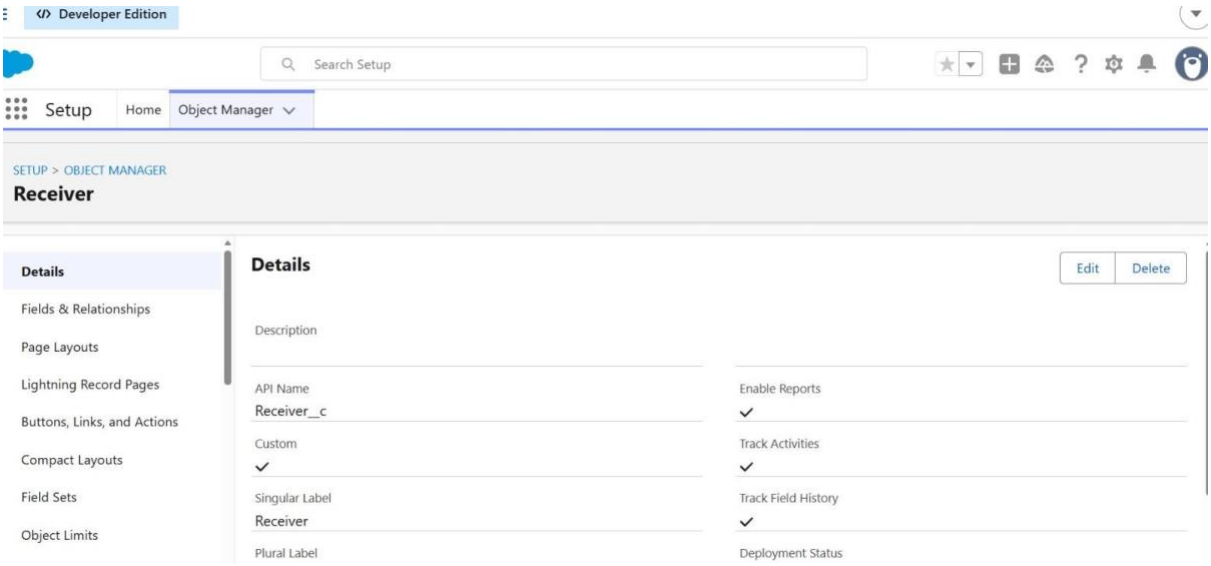
Creating Receiver Object To

create an object:

1. From the setup page
2. Click on Object Manager
3. Click on Create >> Click on Custom Object.
4. Enter the label name as Receiver
5. Enter Plural label name as Receiver's
6. Enter Record Name as ReceiverName 7. Select Data Type as Text:

ReceiverName

8. Select Allow reports.
9. Select Allow search.
10. Allow Track Field History
11. Click on Save and New



Milestone 3 - Tabs

Creating a tab for Parecel Object

1. Go to the setup page → type Tabs in Quick Find bar
2. Click on tabs
3. Click on New (under custom object tab).
4. Select Object (Parcel) >> Select the tab style
5. Click on Next >> (Add to profiles page) keep it as default >> Click on Next (Add to

Custom App) uncheck the include tab.

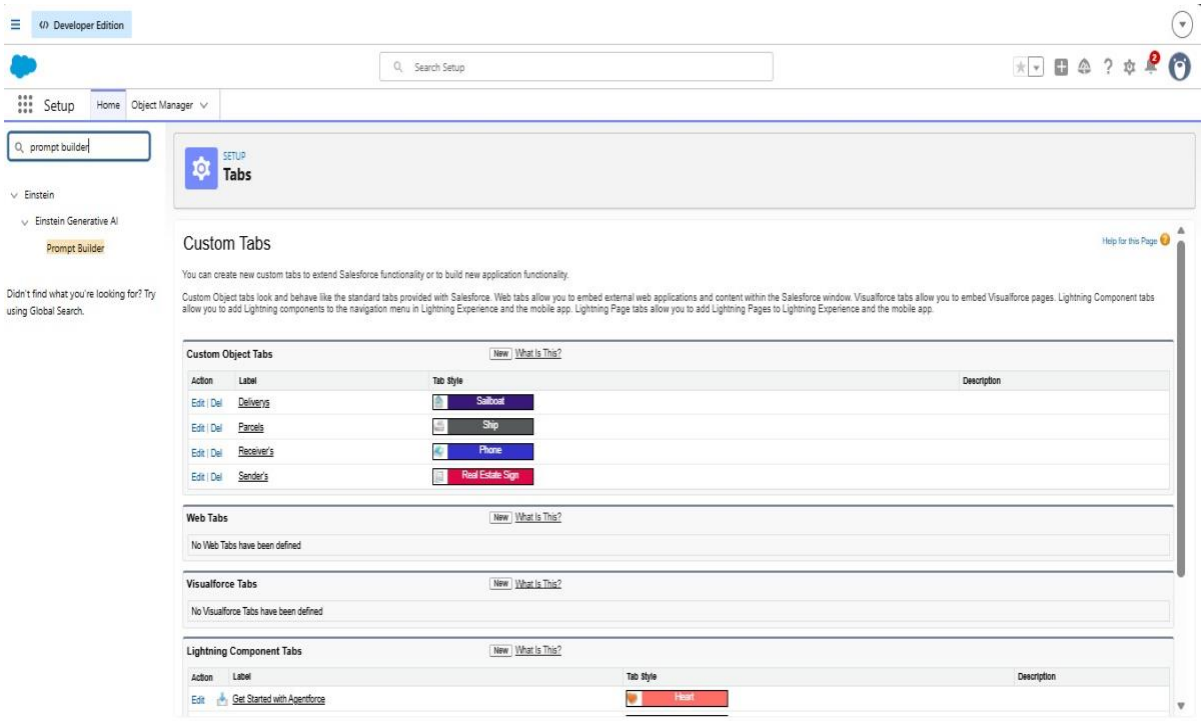
6. Make sure that the Append tab to the user's existing personal customizations is checked.
7. Click save

More Tab Creation

1. Go to Setup → type Tabs in the Quick Find box.
2. Click Tabs under the User Interface section.
3. Click New under Custom Object Tabs.
4. Select the object (Delivery, Sender, or Receiver) → choose a Tab Style.

5. Click Next → keep the Add to Profiles settings as default → click Next.

6. On the Add to Custom Apps page, uncheck “Include Tab.”



7.Ensure Append tab to the user’s existing personal customizations is checked.

8.Click Save and repeat the steps for each object (Delivery, Sender, Receiver).

Milestone 4- Fields & Relationships

Create fields and Relationships for the custom objects created in the previous milestone.

Creating Fields and Relationships for Parcel Object

Beside the Quick Find box you have, Object Manager click on it.

1. Search for and select your Parcel object.

Field 1: Parcel ID

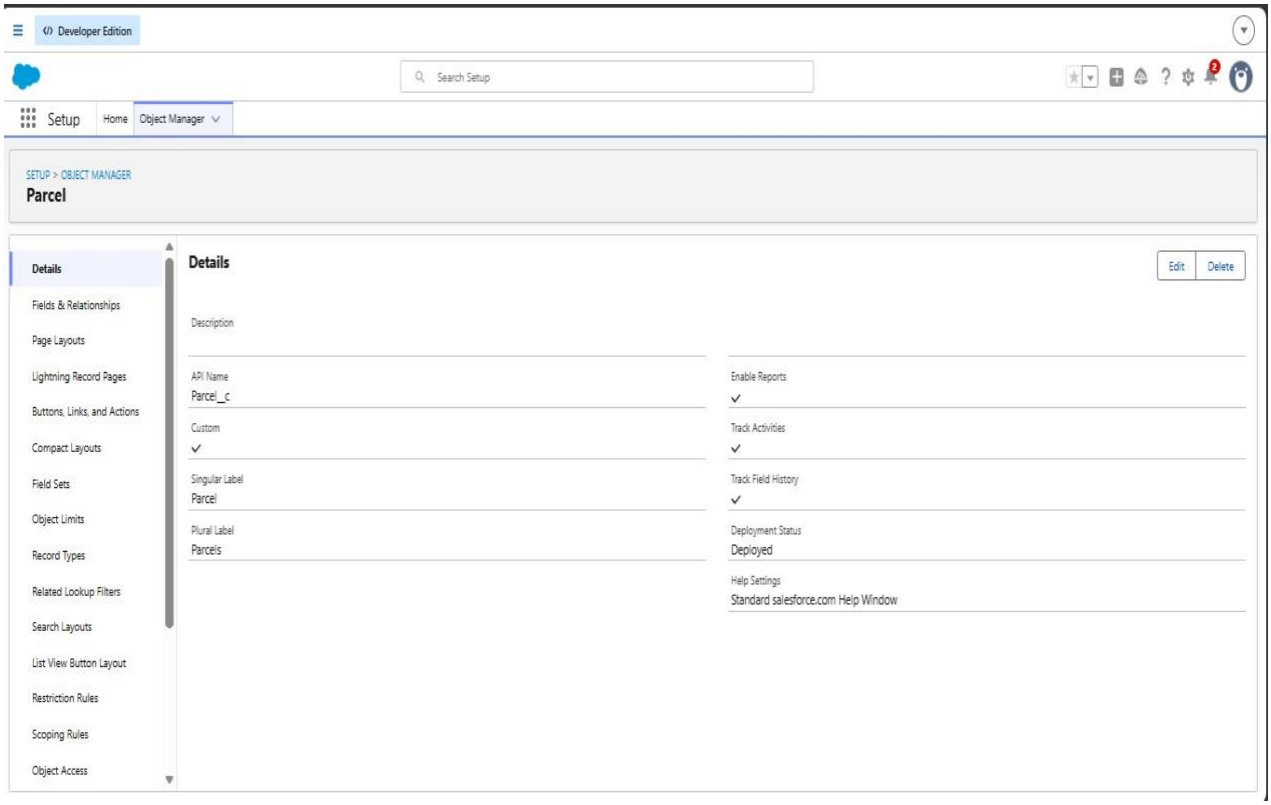
● Go to Fields & Relationships → click New

Enter:

● Data type== auto number→P-{000}

● Click Next → Next → Save & New.

3. Follow step 1 and 2 to create the remaining fields(Status, Weight, Estimated Delivery, Sender) of the Parcel object mentioned in the table below.)



Creating Fields And Relationships For Delivery Object Beside the Quick Find box you have, Object Manager click on it.

1. Search for and select your Delivery object.

Field 1: Current Location

- Go to Fields & Relationships → click New
- Choose Data Type: Geolocation → click Next.

Field Label: Current Location

Click Next → Next → Save & New.

- 3. Follow step 1 and 2 to create the remaining fields(Estimated Delivery, Sender, Parcel) of the Delivery object mentioned in below

Creating Fields And Relationships For Sender Object

Beside the Quick Find box you have, Object Manager click on it.

1. Search for and select your Sender object.

Field 1: Sender Address

- Go to Fields & Relationships → click New
- Choose Data Type: Geolocation → click Next.

Field Label: Sender Address

Click Next → Next → Save & New.

- 3. Follow step 1 and 2 to create the remaining fields (Sender Contact, Sender Email) of the

Sender object mentioned in the table below.

Creating Fields And Relationships For Receiver Object

Beside the Quick Find box you have, Object Manager click on it.

1. Search for and select your Receiver object.

Field 1: Receiver Address

- Go to Fields & Relationships → click New

- Choose Data Type: Geolocation → click Next.

Field Label: Receiver's Address

Click Next → Next → Save & New.

- 3. Follow step 1 and 2 to create the remaining fields (Receiver Contact, Receiver

Email, Sender, Parcel) of the Receiver object mentioned in the table below.

Milestone 5 - Prompt Builder

Designing Prompts for Agent Building.

In Salesforce, go to Setup.

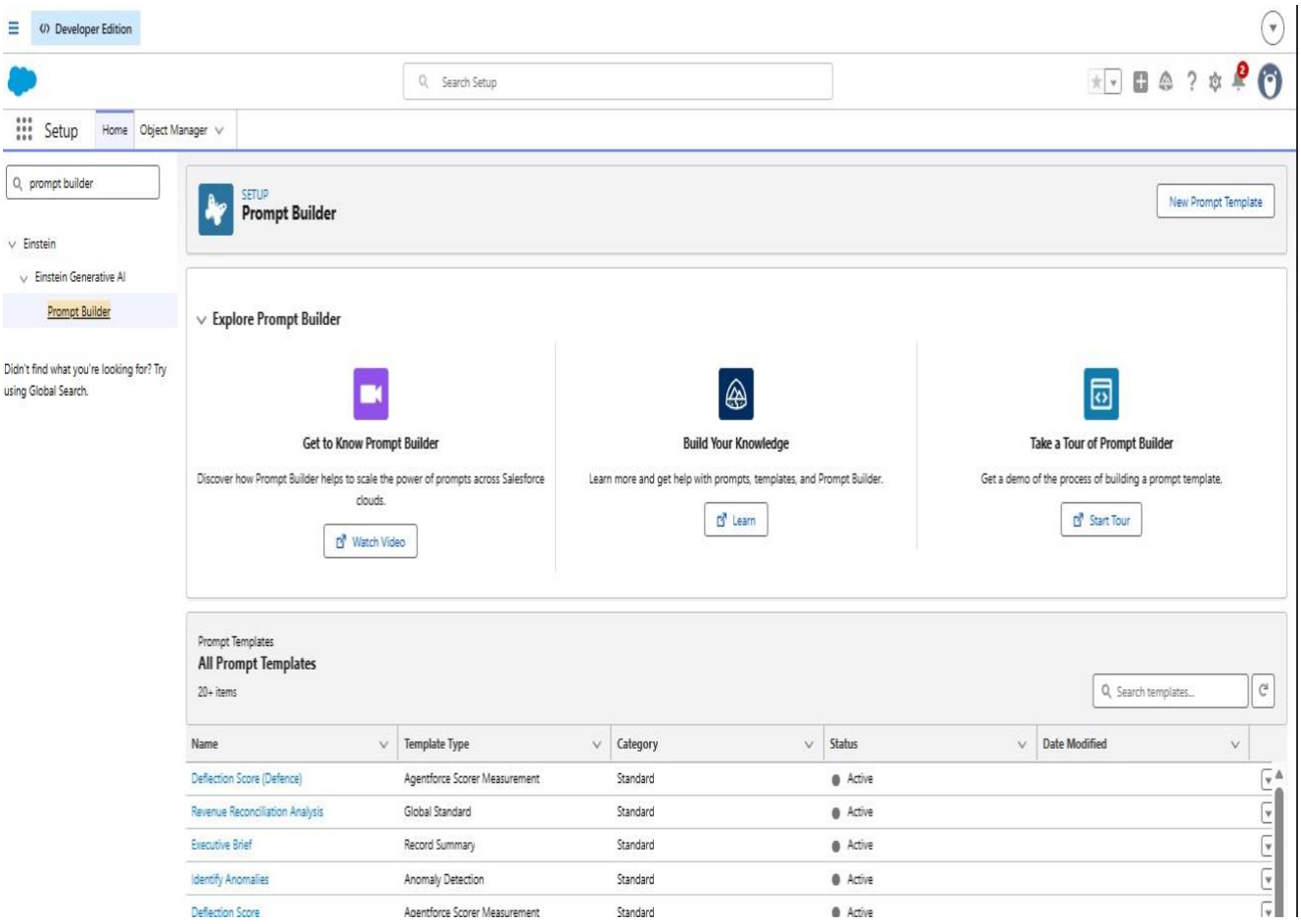
1. In the Quick Find box, type Prompt Templates (or in Agent Console, go to Agent Builder →

Prompt Templates).

2. Click New Prompt Template.-- Record summary template

4. Enter the details

- Label: Retrieve Parcel Details
- API Name: Retrieve_Parcel_Details
- Description: Retrieves parcel details where Parcel_ID matches the user input.



5. Enter the Prompt Template Content

In the Prompt Template Text area, paste the following exactly:

Retrieve details from Parcel c where Parcel_ID__c matches user input. Return response in this format:

Parcel Tracking Update

Parcel Tracking Update

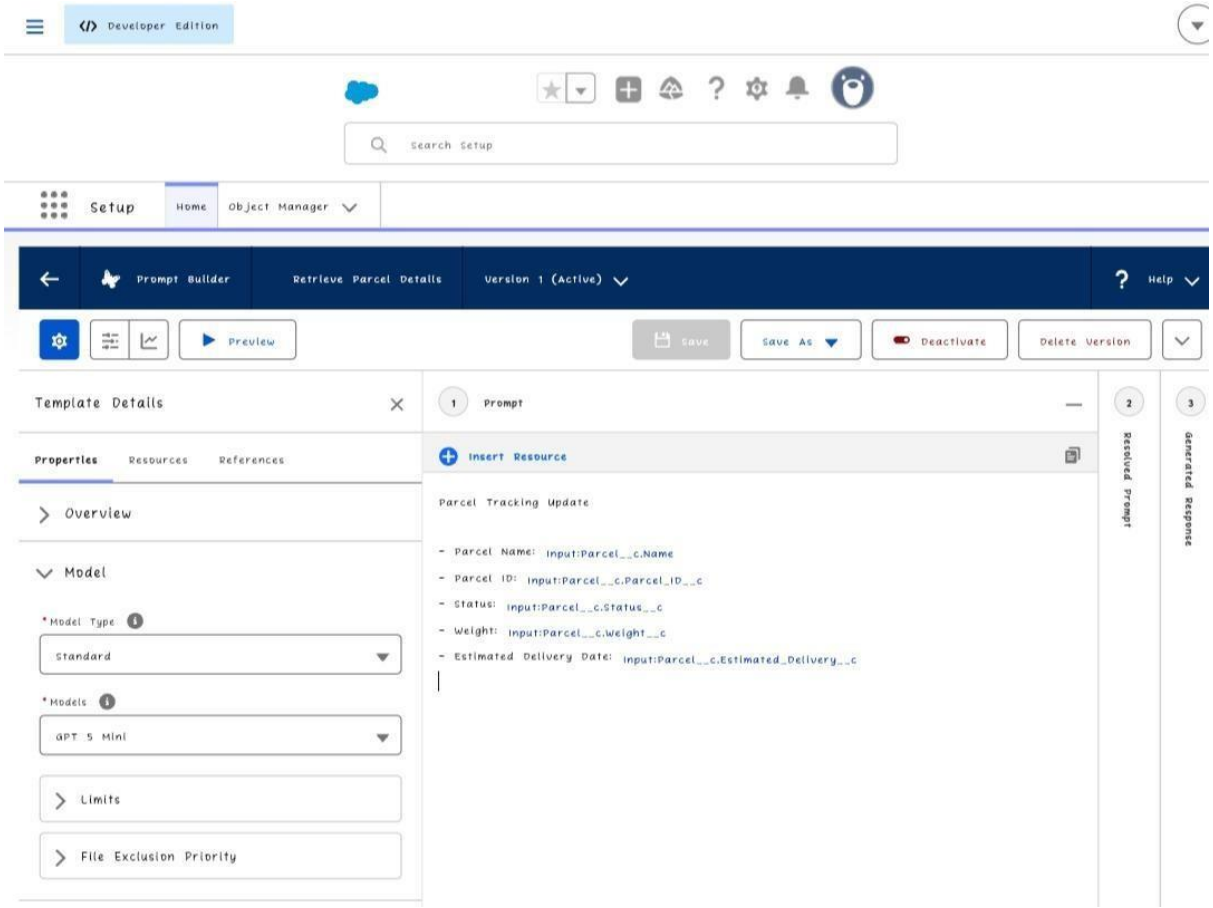
- Parcel Name:{!$Input:Parcel__c.Name}

- Parcel ID:{!$Input:Parcel__c.Parcel_ID__c}

- Status:{!$Input:Parcel__c.Status__c}

- Weight:{!$Input:Parcel__c.Weight__c}

- Estimated Delivery Date:{!$Input:Parcel__c.Estimated_Delivery_Date__c}



6. Add Resource Mapping

7. Scroll down to the Resources / Insert Resources section.

● Click Insert Resource → choose Object.

● Select Parcel__c.

● Add the following fields:

● Name

● Parcel_ID__c

● Status__c

● Weight__c

● Estimated_Delivery_Date__c

8. Click Insert Resource (or Add) for each one.

9. Click Save.

10. Then click Activate to make this Prompt Template available for your agent or flow.

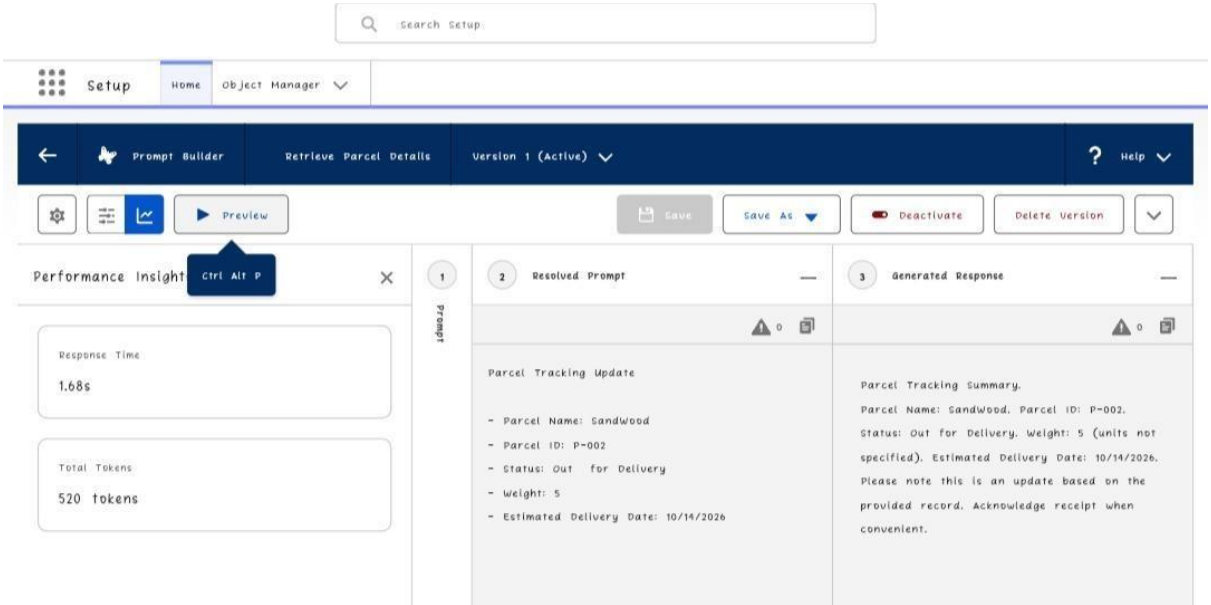
Testing the Prompt

Test the Prompt:

1. Open your Agent preview.

2. Type: “Track parcel P-001”

3. It should return details from the Parcel record.



Milestone 6 - Flows

Create flows for capturing the response as shown.

Go to Setup → search Flows → click New Flow.

1. Choose Auto-Launched Flow (No Trigger) → click Create

2. Add Input Variables (to allow Agent to pass values):

● Click On New Recourse →Variable → Available for Input.

● Ids → Data Type: Text → Available for Input.

● Output → Data Type: Text → Available for Output.

3.Click on the Add icon to add Element: Get Records→ Label: Get Parcel Records

● Object: Parcel

● Field = Parcel_ID__c

● Operator = Equals

● Value = ids

4.Element : Action

1. Find Retrieve parcel details 2.

Click Retrieve parcel details

● Label: Parcel Updates.

● Object To Summarize : Parcel from get Parcel records.

5.Element : Assignment

● Label: Assignment Outputs.

● Variable= Output

● Operator = Equals

● Value = Outputs from Action>Prompt Response

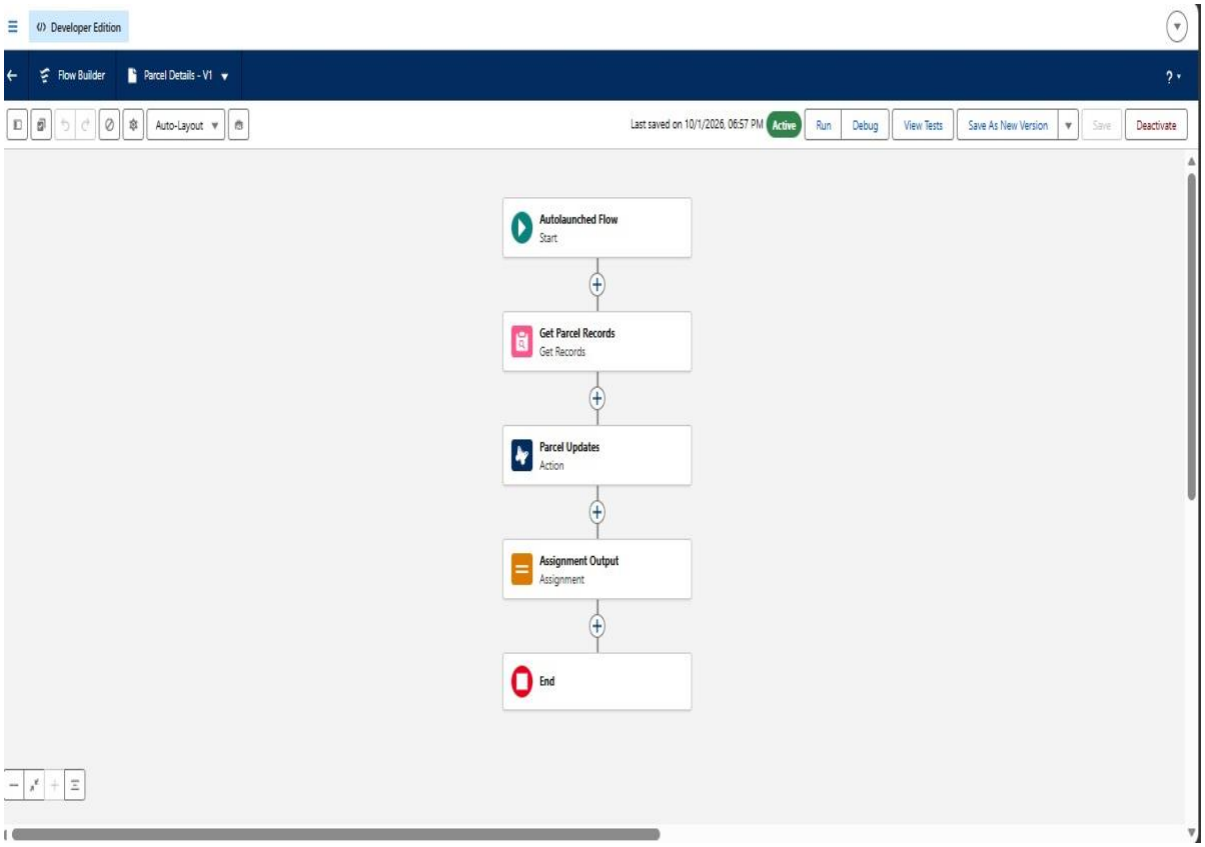
6.Connect Start → Get Records → Action → Assignment → End.

7.Save Flow → Name: Parcel Details.

8.Activate Flow.

9.Once the flow is saved go back to the flows, Click on the drop down near the flow that is just created, click on edit access.

10.On the top look for flow user access and then assign these available profiles, and click save.



Phase 3: UI/UX Development & Customization

To enable agents to efficiently manage parcel operations and ensure secure access control using permission sets and AI-driven agent experiences built with AgentForce.

Milestone - 1 Creating Permission Sets

In the Quick Find box (left side), type Permission Sets.

1. Click Permission Sets under Users.

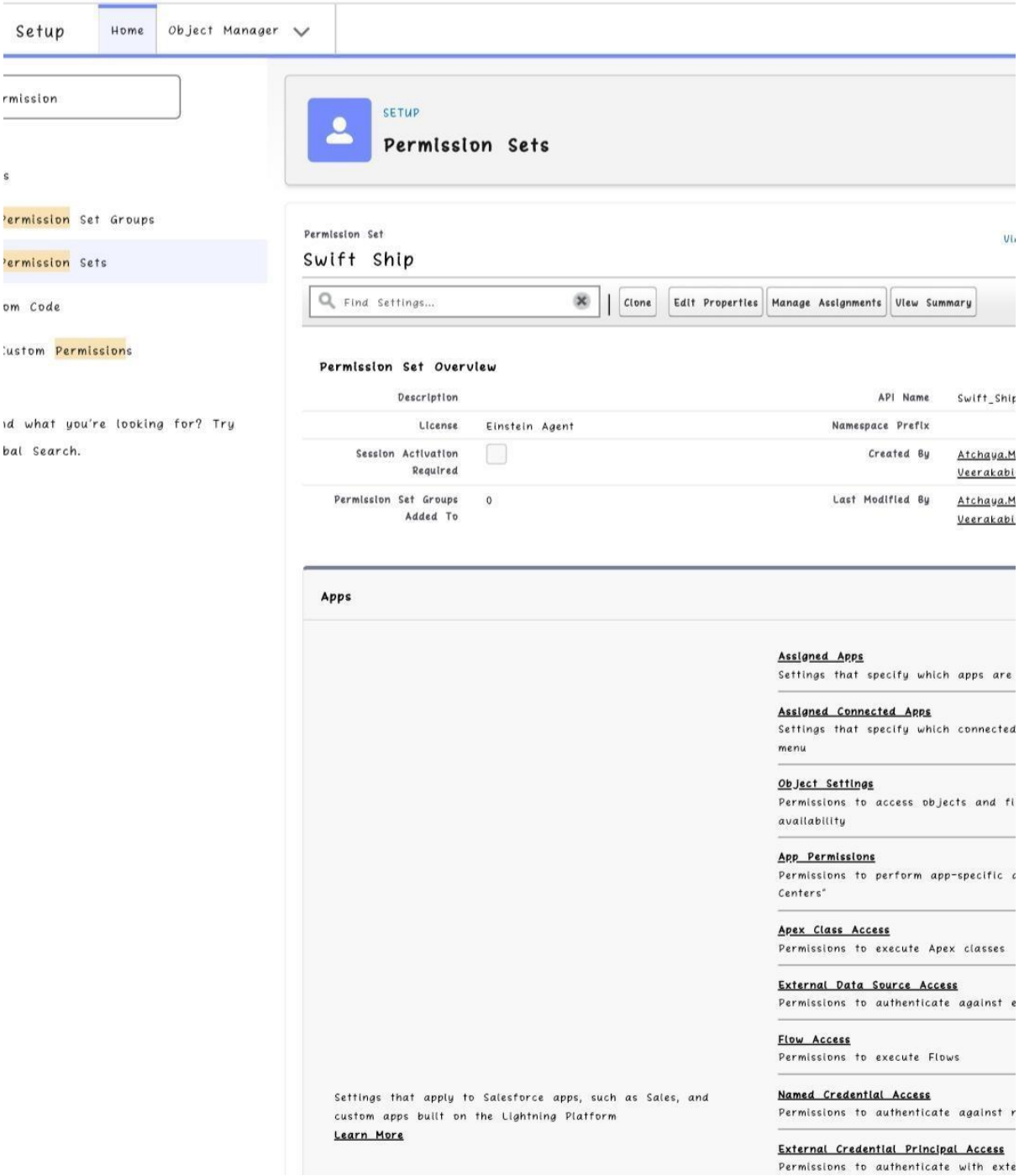
1. Click the New button.

2. Enter the following details:

- Label: Swift Ship
- API Name: auto-fills
- License:

Choose Einstein Agent.

3. Click Save.



Inside the Permission Set page → under Apps → Object Settings, click Object Settings.

1. Search and click each object one by one

2. Click Edit.

Enable the following:

- Read
- Create

- Edit

Under Field Permissions, ensure at least Visible is checked for all key fields.

Click Save after each object.

- ✞ Parcel _____ c
- ✞ Delivery _____ c
- ✞ Sender✞ Receiver__c _____ c

Milestone - 2 Assigning Permissions to your Agent user

In the Quick Find box on the left, type Users.

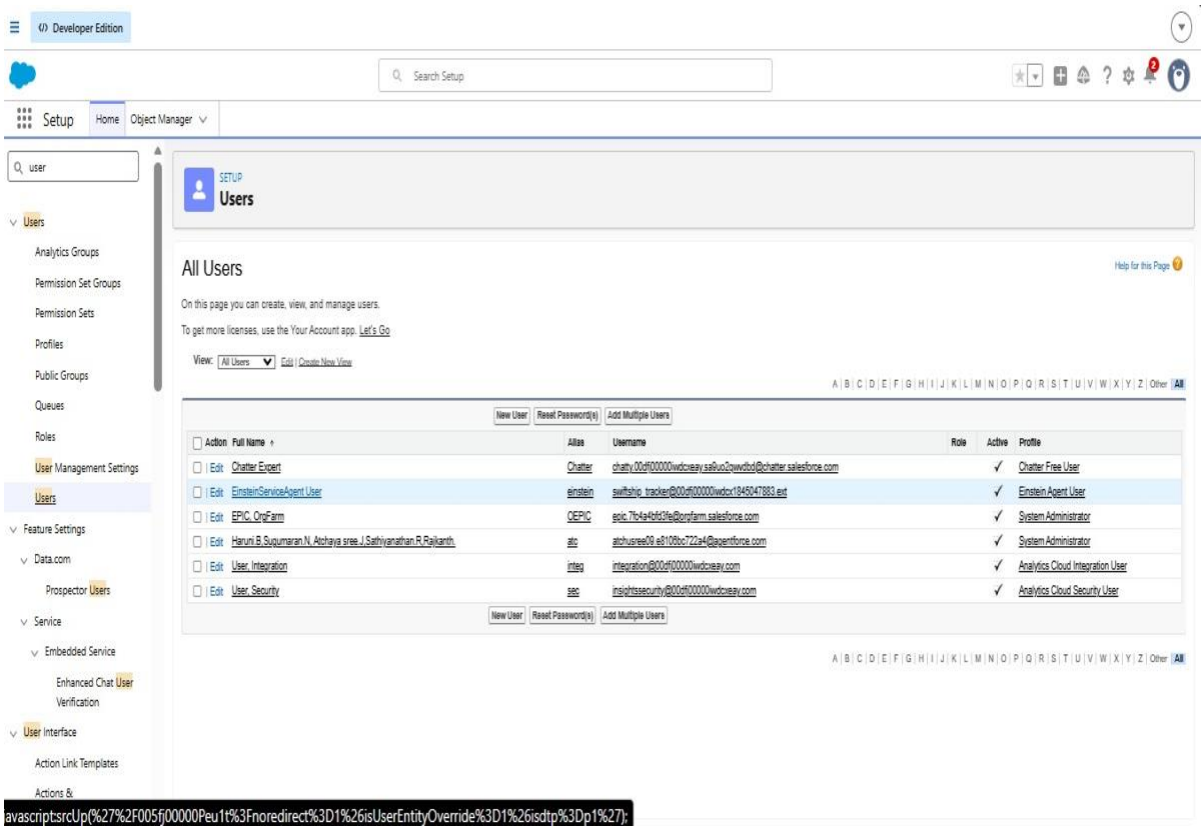
1. Click Users under the Users section.

2. Open the Agent User Record

From the list of users, find the Einstein Service Agent user to represent your Agent

(e.g., “SwiftShip Agent”).

Click on the Full Name of that user to open the record.



3.Scroll down to the section named Permission Set Assignments.

4. Click the Edit Assignments button.

1. On the left side (Available Permission Sets), find and select your permission set:

● SwiftShip

Click the Add → button to move it to the right-side list (Enabled Permission Sets).

Click Save.

Milestone -3 Org Setup for AgentForce

Enable salesforce Default agent from Salesforce Go

1. From Quick find box→ search “Salesforce Go” → Click on it

2. Search for Default agent —> click get started→ and turn on

3. And refresh the page

Enable Einstein Setup

Go to Setup → Quick Find → Einstein Setup.

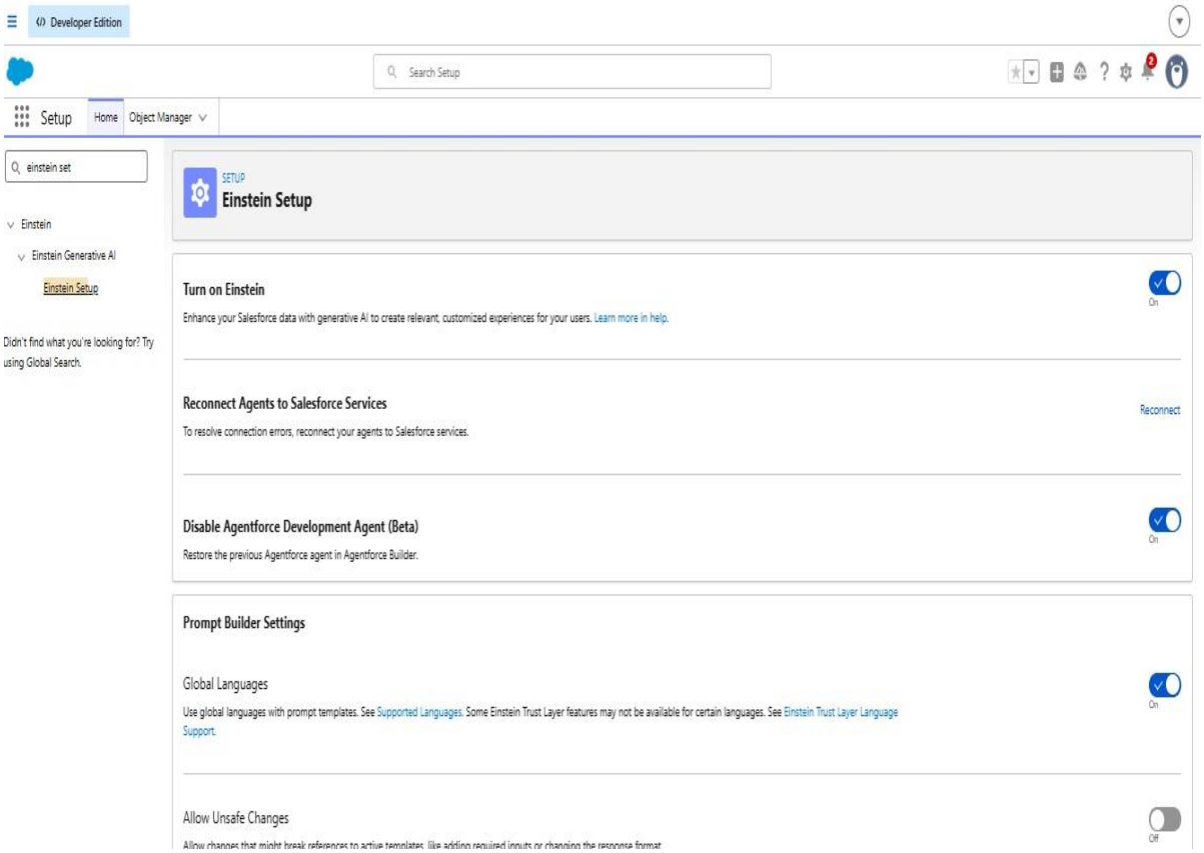
1. Open Einstein Setup → click Get Started if prompted.

Toggle On these key services:

● Turn On Einstein

● Global Languages

● Deploy Prompt Template

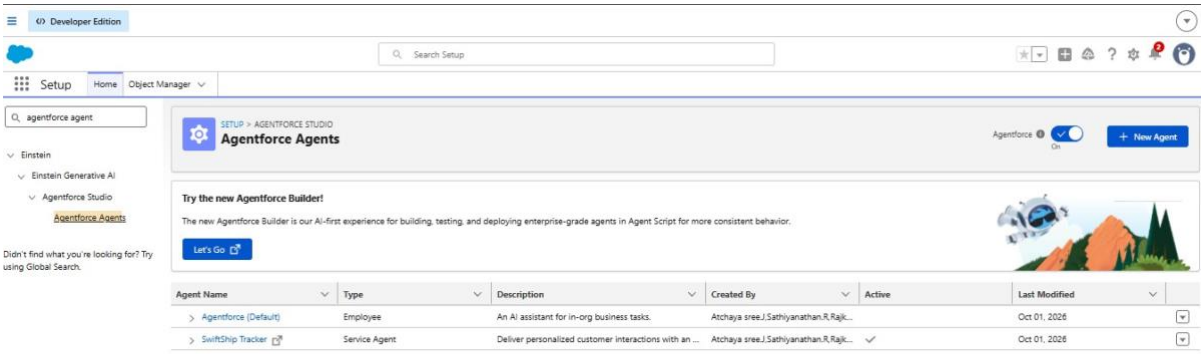


3.Refresh your browser page.

3. Go to Setup → Quick Find → AgentForce Agents.

Toggle On these key services:

- Agentforce
- Enable Agentforce Default Agent



Milestone - 4 Building The Agent

1. In Setup, search for Agentforce Agents in Quick Find.

2. Click Agentforce Agents, then click on the default Agent, Then click on Open in Builder.

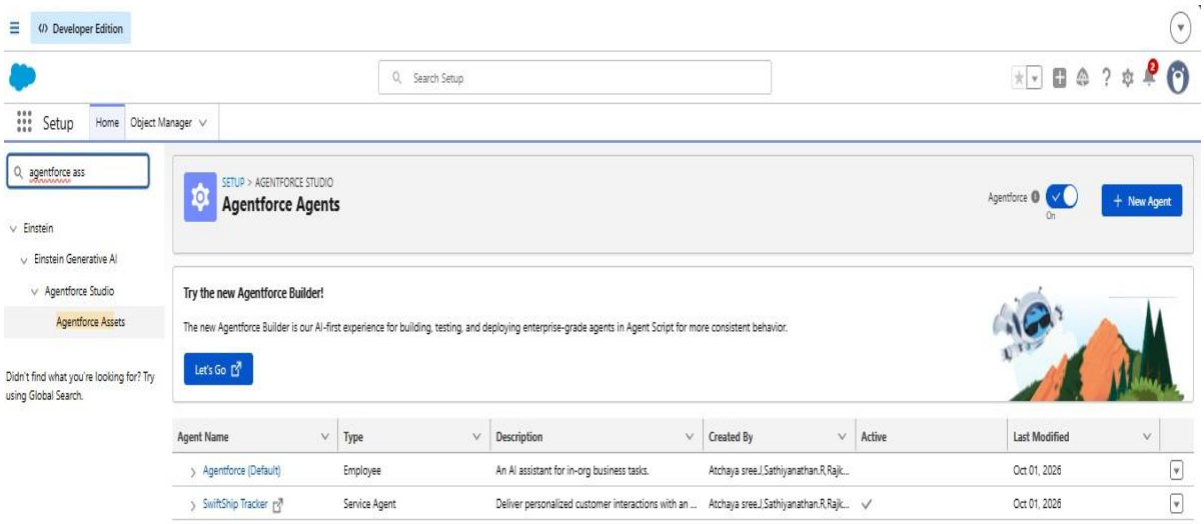
Assign Sub Agents to the Agent On

the subagents Click on New.

1. Click New Subagent

2.Create a new subagent.

3.Fill in the details as shown.



Adding Actions to Agentforce Assets

While still on the Agent Builder in the topics, Click on the Topics Actions

1.Click on New Agent Action.

2.For Action type, Click on Flow then Next.

3.For Reference Flow use the recently created flow, click Next.

4.Fill the details for Agent Configurations.

Assign Subagents Action

While on the Agent Builder in Subagent, Click on the Parcel Updates Subagent

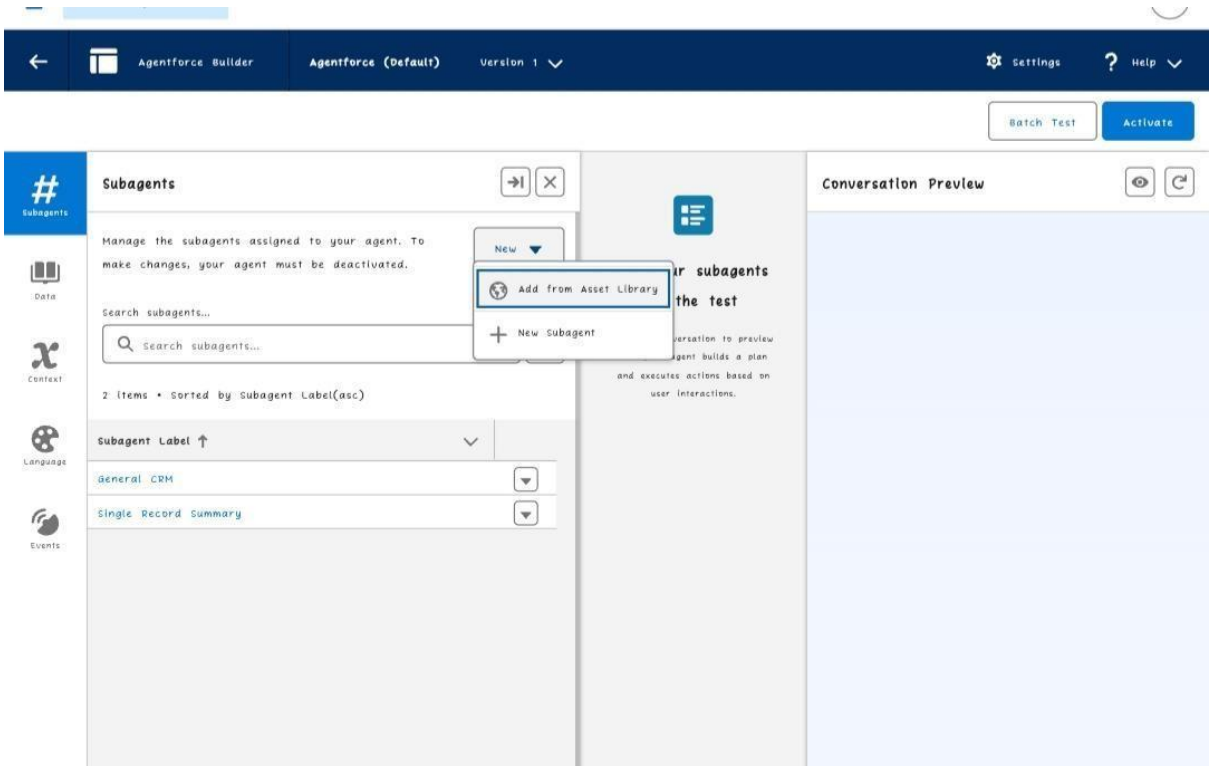
1.Click on the subagents Actions, then select Add from asset library.

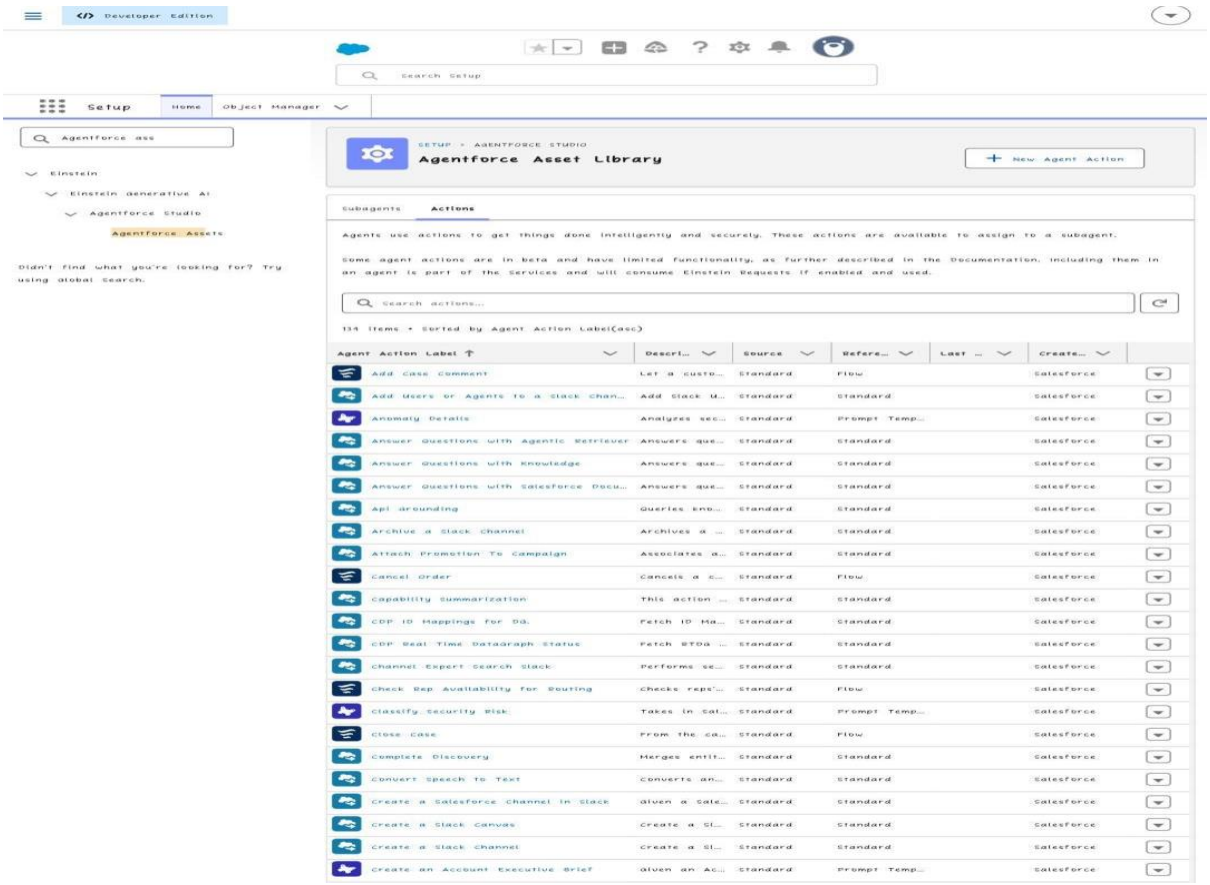
2.Select the flow action that we created and click on finish.

3.Click on settings , and rename your agent to SwiftShip Tracker

4.Click Save.

5.Click on Ignore and Activate, then Activate.





Phase 4: Data Migration, Testing & Security

To validate the functionality and reliability of the Parcel Tracking Agent created in Agent Builder. This

phase ensures that the agent correctly recognizes topics, triggers the assigned Flow (SwiftShip

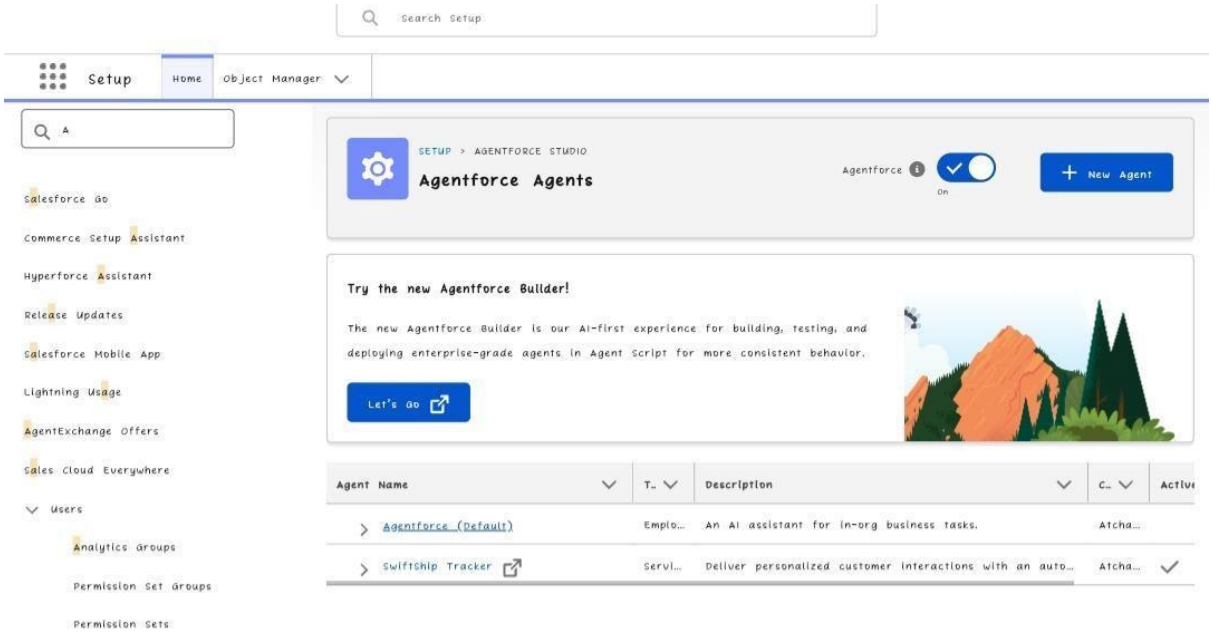
Tracker), retrieves or updates parcel data, and performs all intended actions without access or

permission issues. The goal is to confirm that the AI-driven agent operates seamlessly within

Salesforce and provides accurate, real-time responses to parcel management queries.

Verify Agent

Go to Setup → Agentforce Agents.



1. Open the Agents list and confirm that your agent

Name: Parcel Tracking Agent, appears

Validate Assigned Topic

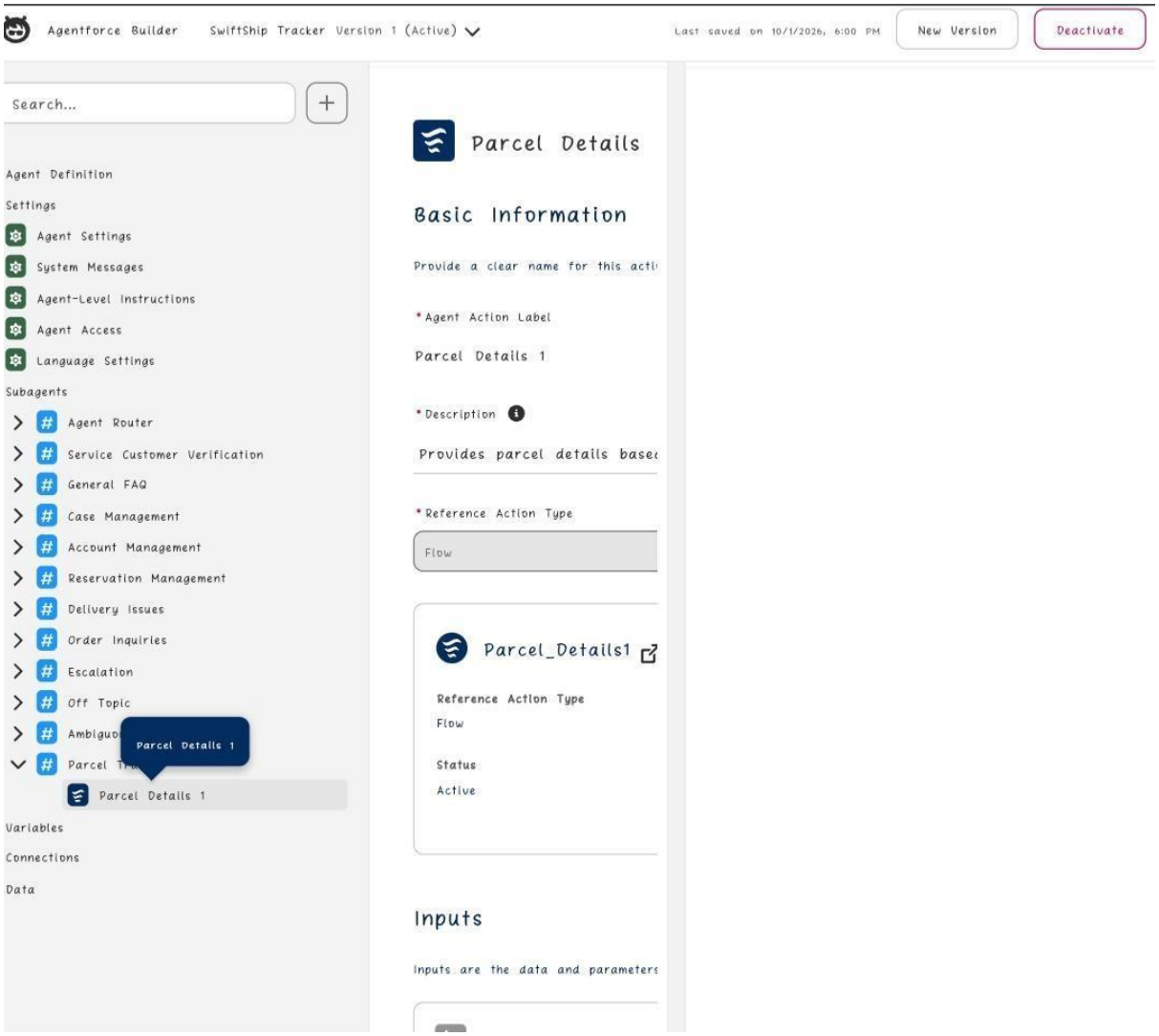
In Agent Builder → SubAgent, confirm that the SubAgent you created (e.g., Parcel Tracking or

SwiftShip Tracker) is listed.

Open the topic to verify:

- It contains a prompt or description related to parcel management.
- The Action/Flow (SwiftShip Tracker) is linked to this topic.

2.Click Save and ensure the topic is Active.



Check the Flow Association

Go to Setup → Flows.

1. Locate the Flow named SwiftShip Tracker.

2. Confirm:

- The Flow is Activated.
- The Flow includes screens or actions that handle:
- Parcel ID input
- Parcel details retrieval
- Status update or confirmation message

3. Open the Flow's Advanced Settings → ensure Run in System Context (Without Sharing) if data access is restricted.

Activate and Test the Agent

1. Return to Setup → Agent Builder → Parcel Tracking Agent.

2. Click Activate (if not already activated).

3. Once active, test using Preview / Try it Out mode:

Type a query such as:

"Track parcel ID P-001" or

"Update delivery status for parcel P-002 to Delivered."

4. Observe the agent's response.

Validate Data in Salesforce

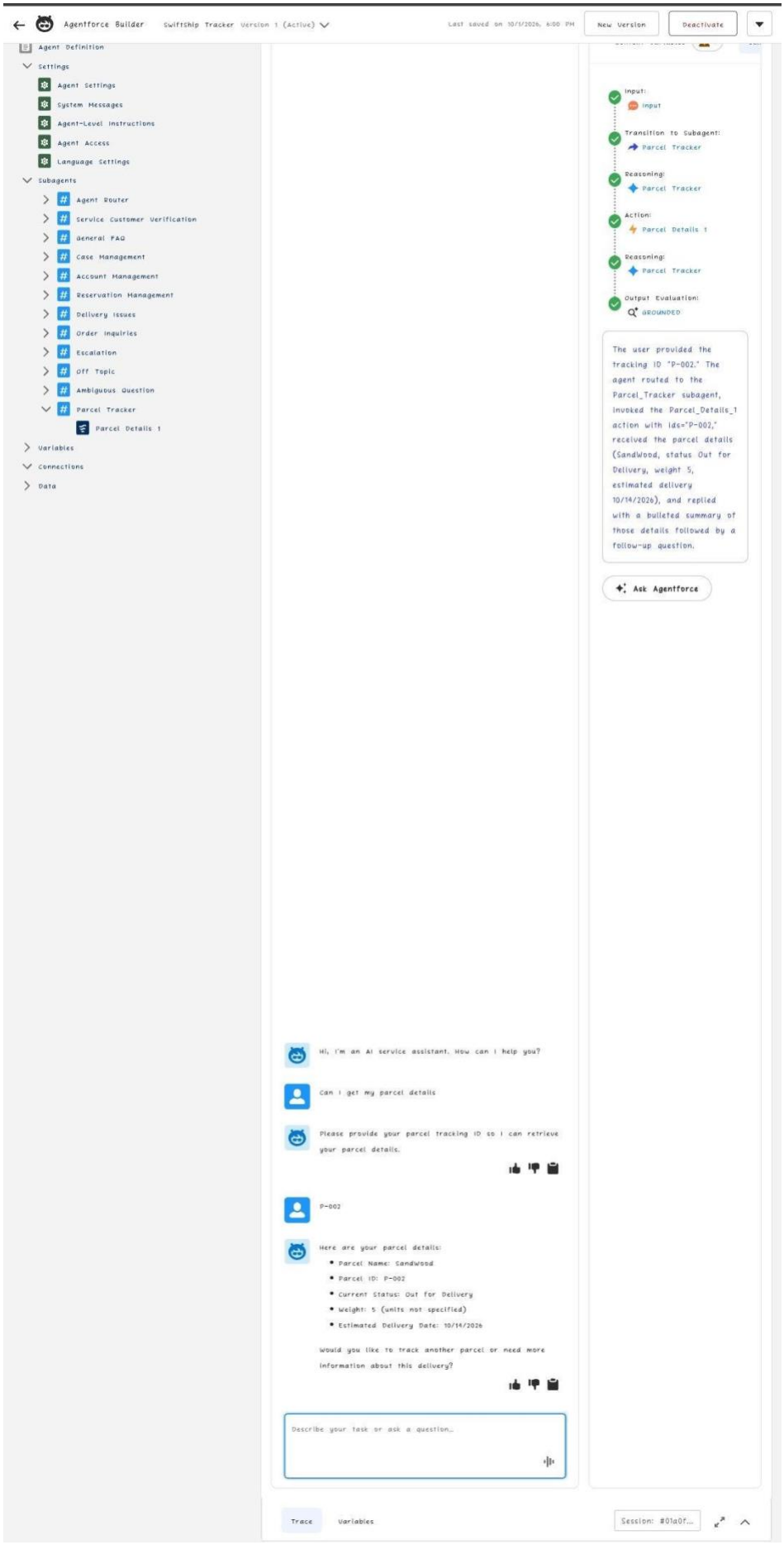
1. After the agent runs the Flow:

- Go to App Launcher → Parcels (Parcel__c).

Search for the Parcel ID you used during testing.

Verify:

- The Status or related field is updated.
- Any Flow-created records (bookings, updates) appear correctly.


Agentforce Builder
SwiftShip Tracker Version 1 (Active)
Last saved on 10/1/2026, 6:00 PM
New Version
Deactivate
Agent Definition
Settings
Agent Settings
System Messages
Agent-Level Instructions
Agent Access
Language Settings
Subagents
Agent Router
Service Customer Verification
General FAQ
Case Management
Account Management
Reservation Management
Delivery Issues
Order Inquiries
Escalation
Off Topic
Ambiguous Question
Parcel Tracker
Parcel Details 1
Variables
Connections
Data
Input:
Input
Transition to Subagent:
Parcel Tracker
Reasoning:
Parcel Tracker
Action:
Parcel Details 1
Reasoning:
Parcel Tracker
Output Evaluation:
GROUNDED
The user provided the tracking ID "P-002." The agent routed to the Parcel_Tracker subagent, invoked the Parcel_Details_1 action with id="P-002," received the parcel details (SandWood, status Out for Delivery, weight 5, estimated delivery 10/14/2026), and replied with a bulleted summary of those details followed by a follow-up question.
Ask Agentforce
Hi, I'm an AI service assistant. How can I help you?
Can I get my parcel details
Please provide your parcel tracking ID so I can retrieve your parcel details.
P-002
Here are your parcel details:
Parcel Name: Sandwood
Parcel ID: P-002
Current Status: Out for Delivery
Weight: 5 (units not specified)
Estimated Delivery Date: 10/14/2026
Would you like to track another parcel or need more information about this delivery?
Describe your task or ask a question...
Trace
Variables
Session: #01a0f...

Phase 5: Deployment, Documentation & Maintenance

- In the scope of our SwiftShip Tracker (AgentForce) project implementation:
- The purpose of this phase is to validate that the Parcel Tracking Agent, created using Salesforce

Agent Builder, works as intended within the Salesforce Developer Org environment.

- The main goal is to understand how an AI-powered Service Agent can be tested to ensure correct topic recognition, Flow execution, and data retrieval from Salesforce custom objects.
- Testing is conducted within the Developer Org to simulate how the Agent would perform in a live

setup, ensuring that permissions, flows, and interactions function correctly before any deployment

stage.

- Real-world production testing (UAT or sandbox deployment) is not performed in this project because:
- Developer Edition orgs are standalone environments and do not support productionlinked testing environments.
- Enterprise-level AgentForce deployment involves sandbox testing, staging environments, and CI/CD automation, which are outside the scope of this academic/prototype implementation.

Maintenance, Monitoring & Troubleshooting

In the SwiftShip Tracker (AgentForce) project, ongoing maintenance ensures that Parcel, Customer,

and Delivery data remain accurate, synchronized, and up to date. Regular monitoring helps detect issues such as misrouted agent queries, failed Flow executions, or performance delays in

AgentForce responses. Troubleshooting activities include resolving configuration errors, updating

Flow logic, refining agent topics, and improving prompts to enhance user interactions.

Although this project is implemented within a Salesforce Developer Org for learning purposes, in

real-world enterprise deployments, these maintenance and monitoring practices are essential. They

ensure that AI-powered service agents remain reliable, scalable, and continuously aligned with evolving customer service operations and business goals.

7. FUNCTIONAL & PERFORMANCE TESTING

During testing, screenshots were captured from Salesforce to validate vehicle creation, dealer assignment, stock validation triggers, order status updates, batch processing, scheduled reminders, and dashboard analytics. All features were verified to ensure correct functionality and performance.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralized parcel and customer information in Salesforce.
- Provides better visibility into parcel booking and delivery tracking.
- Automates key logistics processes using Salesforce Flow.
- Agentforce enables conversational customer support and parcel-status assistance.
- Prompt Builder supports AI-powered responses and self-service.
- Reduces manual effort for customers, support teams, and administrators.
- Provides timely notifications and updates during the delivery process. ●

Experience Cloud can provide customers with convenient access to tracking information.

- Supports integration with external logistics systems.
- Can scale to support multiple locations and growing parcel operations.

Disadvantages

- Requires accurate and up-to-date parcel and customer data.

- Dependence on Salesforce configuration and automation.

- Agentforce responses depend on properly configured prompts and available data.
- External logistics integrations may increase implementation complexity. ● Flow and automation logic require regular maintenance when business processes change.
- Users need appropriate Salesforce or Experience Cloud access to use the system.
- Initial Salesforce implementation and configuration can require significant effort.

● Expanding the system to additional locations or integrations may require further customization.

9. Conclusion:

The SwiftShip Tracker demonstrates how Salesforce can be used to build a centralized parcel management and tracking solution. The project integrates custom objects, relationships, Flows, Apex, Email Alerts, Experience Cloud, AgentForce AI, Prompt Builder, security controls, and reporting to support parcel booking, delivery tracking, status updates, and customer communication.

The integration of AgentForce AI with Prompt Builder and Salesforce Flow provides a conversational approach to parcel tracking, enabling customers to obtain shipment information without depending entirely on manual support. The project therefore provides a practical foundation for a scalable and user-friendly parcel management solution.

10. FUTURE SCOPE

● Extend AgentForce capabilities for more parcel-management queries.

● Enhance real-time delivery tracking.

● Expand customer self-service through Experience Cloud.

● Add more advanced delivery-performance analytics.

- Integrate additional external logistics systems.
- Improve AI prompts and conversational experiences.

- Introduce enterprise-level sandbox and CI/CD deployment.
- Expand automation for additional parcel operations.

The source identifies enterprise deployment, continuous monitoring, prompt refinement, Flow improvement, and AI-agent scalability as areas for future enhancement.

11. APPENDIX

- A. Custom Objects: Parcel, Delivery, Sender, Receiver
- B. Standard Objects: Account, Contact, Case, Task, EmailMessage, User
- C. Automation: Flows, Approval Processes, Email Alerts
- D. Apex: Apex Classes, Batch Apex
- E. AI: AgentForce AI, Prompt Builder
- F. UI: Lightning App, Lightning Record Pages, Experience Cloud
- G. Security: Profiles, Roles, Permission Sets, Sharing Rules, Field-Level Security
- H. Analytics: Reports & Dashboards
- I. Testing: Flow, AgentForce, data retrieval, security and permission validation

Thank You